

■■ NewsGenie — AI-Powered Information & News Assistant

NewsGenie is an agentic AI application that delivers real-time, curated news and answers general queries through a unified conversational interface. Built with **LangGraph**, **OpenAI GPT-4o**, **NewsAPI**, and **Streamlit**.

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Overview

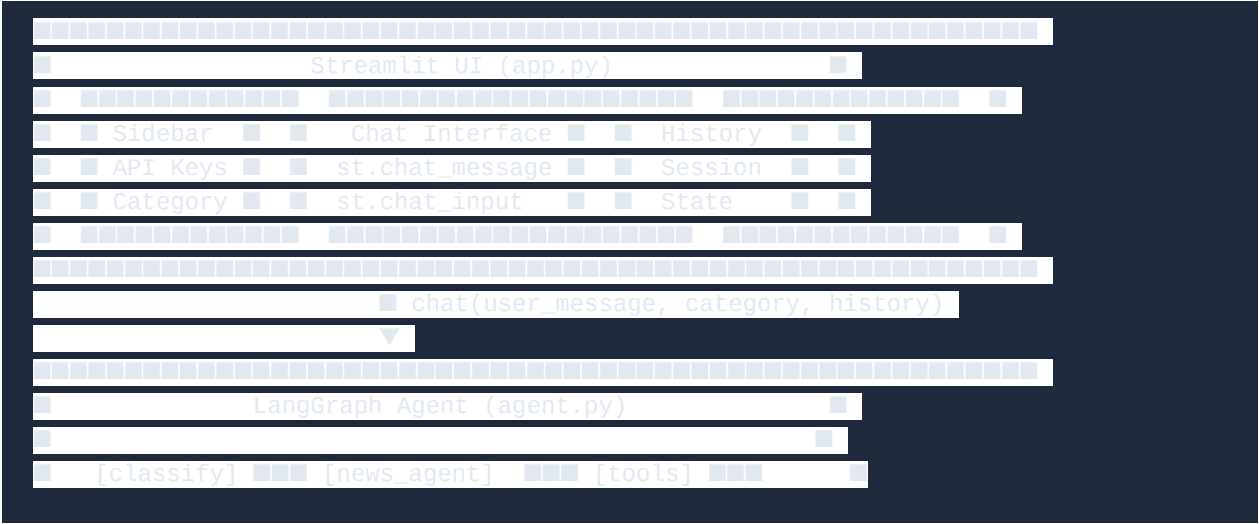
People struggle to keep up with real-time, reliable news in today's fast-paced world. NewsGenie solves this by combining:

- An **AI chatbot** that distinguishes news queries from general questions
 - A **real-time NewsAPI integration** for structured, sourced headlines
 - A **DuckDuckGo web search fallback** that keeps the app functional even without a NewsAPI key
 - A **LangGraph-based agentic workflow** that routes, fetches, and synthesises information intelligently
 - A **Streamlit chat interface** with session memory, category selection, and graceful error handling
-

Features

Feature	Description
Query classification	GPT-4o-mini automatically decides if a query is news-related or general
7 news categories	Technology, Finance, Sports, Health, Science, Entertainment, General
Real-time headlines	Top-5 articles per category via NewsAPI with source & date attribution
Keyword news search	Search any topic across all NewsAPI sources
Web search fallback	DuckDuckGo search activates automatically when NewsAPI is unavailable
Conversation memory	Full multi-turn context preserved across the session
Streamlit chat UI	Native <code>st.chat_message</code> interface with real-time agent status
Session management	Clear conversation button resets both display and LangChain history
Error handling	Distinct messages for auth errors, rate limits, and missing packages

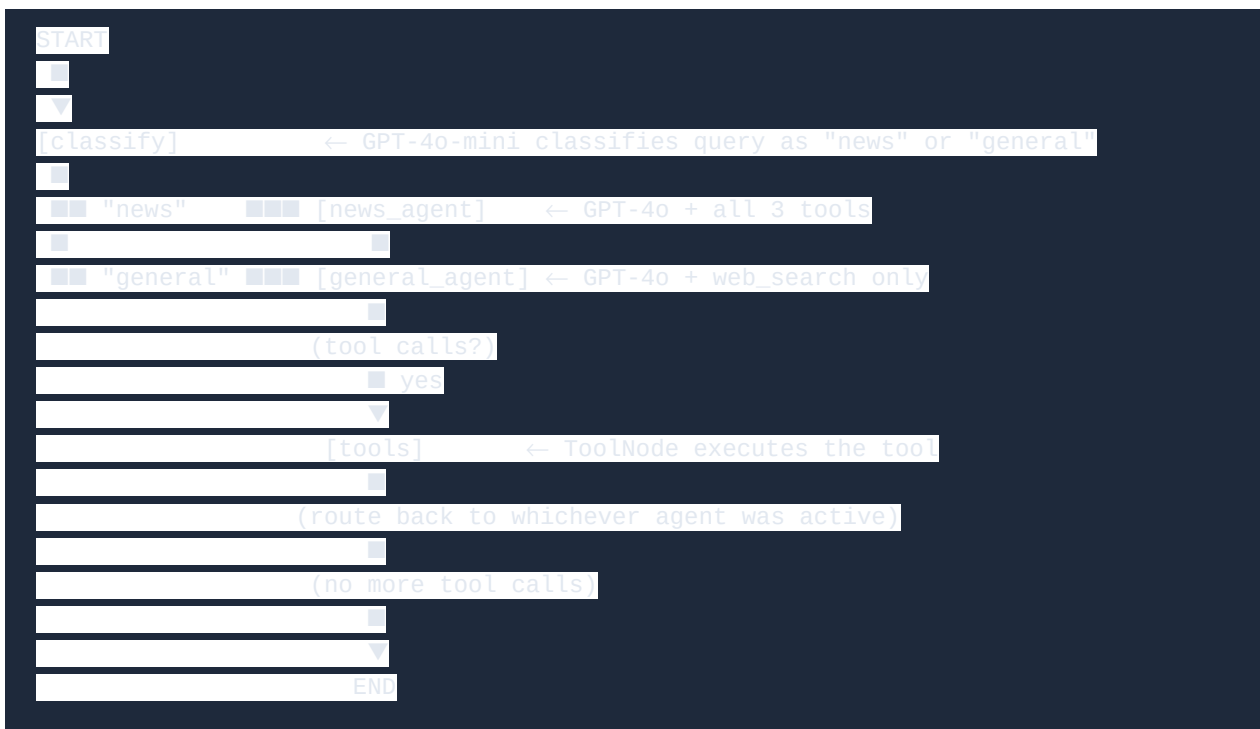
Architecture





LangGraph Workflow

The agent is built as a stateful directed graph with four nodes:



State Schema



Routing Logic

- **After classify** → `news_agent` if `query_type == "news"`, else `general_agent`
- **After tools** → back to whichever agent set `active_agent` (enables multi-step tool use)
- **After agent** → `tools` if tool calls present, else `END`

Tools

`get_top_headlines(category)`

Fetches the top 5 headlines for a given category.

- **Primary:** NewsAPI `/v2/top-headlines?category=<category>`
- **Fallback:** DuckDuckGo search for `"latest <category> news today"`
- Returns: title, source, publication date, description, article URL

`search_news_by_keyword(query)`

Searches all NewsAPI sources for articles matching a keyword or phrase.

- **Primary:** NewsAPI `/v2/everything?q=<query>&sortBy=publishedAt`
- **Fallback:** DuckDuckGo search
- Returns: top 5 matching articles with metadata

`web_search(query)`

General-purpose web search for supplementary context or general queries.

- **Source:** DuckDuckGo (via `langchain-community`)
- Used by both agents; sole tool available to the general agent

Project Structure

```
newsgenie/  
├── app.py           # Streamlit UI – chat interface, sidebar, session management  
├── agent.py         # LangGraph agent – classify, route, news/general nodes  
├── tools.py         # Tool definitions – NewsAPI + DuckDuckGo with fallbacks  
├── requirements.txt # Python dependencies  
├── .env             # API keys (not committed to version control)  
└── venv/            # Python virtual environment
```

Setup & Installation

Prerequisites

- Python 3.8 or higher
- An **OpenAI API key** — platform.openai.com/api-keys
- A **NewsAPI key** (optional but recommended) — free tier at newsapi.org

1. Clone / navigate to the project

```
cd newsgenie
```

2. Create and activate virtual environment

```
# Create
python -m venv venv

# Activate — Windows PowerShell
.\venv\Scripts\Activate.ps1

# Activate — Windows Command Prompt
.\venv\Scripts\activate.bat

# Activate — macOS / Linux
source venv/bin/activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

Configuration

Create a `.env` file in the project root:

```
# Required — OpenAI API key
OPENAI_API_KEY=sk-...

# Optional but recommended — NewsAPI key (free at newsapi.org)
NEWS_API_KEY=your_newsapi_key_here
```

Without a NewsAPI key the app still works — it falls back to DuckDuckGo web search automatically, though results are less structured (no source attribution or article links).

API keys can also be entered directly in the app's sidebar at runtime.

Running the App

```
streamlit run app.py
```

The app opens at **http://localhost:8501** by default.

Usage Guide

News queries

Ask anything that involves current events — the classifier will route it to the news agent, which fetches live data before responding.

```
"Show me the latest technology news"
"What happened in the stock market today?"
"Give me today's top sports headlines"
"What's the latest on AI regulation?"
```

General queries

Any non-news question is routed to the general agent, which uses web search when needed.

```
"What is machine learning?"
"Explain how interest rates affect inflation"
"Who founded OpenAI?"
```

Category selector

Use the **sidebar radio buttons** to set your preferred news category. The active category is passed to the news agent as context, so asking "show me the news" will fetch headlines for your selected category.

Conversation memory

NewsGenie maintains the full conversation history within a session. You can ask follow-up questions naturally:

```
User: "What's happening in AI?"
Genie: [fetches AI news]
User: "Tell me more about the first story"
Genie: [responds with context from previous exchange]
```

Click **■ ■ Clear conversation** in the sidebar to start a fresh session.

Error Handling & Fallbacks

Scenario	Behaviour
NewsAPI key missing	Falls back to DuckDuckGo web search silently
NewsAPI returns no results	Falls back to DuckDuckGo web search
NewsAPI request timeout / error	Falls back to DuckDuckGo web search
DuckDuckGo unavailable	Returns descriptive error message in response

Invalid OpenAI API key	Shows "Invalid or missing OpenAI API key" error banner
OpenAI rate limit hit	Shows "Rate limit hit, please wait" warning banner
Missing Python package	Shows package name and install command
Classifier fails	Defaults to "news" mode to favour retrieval over silence

Tech Stack

Component	Technology
LLM	OpenAI GPT-4o (agents), GPT-4o-mini (classifier)
Agent framework	LangGraph 1.x
LLM integration	LangChain + langchain-openai
News data	NewsAPI.org (/v2/top-headlines, /v2/everything)
Web search	DuckDuckGo via langchain-community + ddgs
Frontend	Streamlit 1.x
Environment	python-dotenv
HTTP client	requests